

0-15

gate, algo, CSE, DA
== < == < ==

Lec_16

0, 0, 5
6, 1, 2

GATE LEVEL QUESTIONS

① State T/F.
if $f(n) = \Omega(g(n))$, than, $2^{f(n)} = \Omega(2^{g(n)}) \Rightarrow$ fail

Solve

$$f(n) = \Omega(g(n))$$

$$\Leftrightarrow f(n) \geq c \cdot g(n)$$

$$2^{f(n)} \geq c \cdot 2^{g(n)}$$

Ex:

① $f(n) = n^2$, $g(n) = n$
 $n^2 \geq n$

$$2^{n^2} > 2^n$$

$$f(n) = n$$

$$n \geq c \cdot 2^n$$

$$n \geq \frac{1}{2}n$$

$$c = \frac{1}{2}$$

fail

$$g(n) = 2^n$$

$$2^{f(n)} \geq c \cdot 2^{g(n)}$$

$$2^n \geq c \cdot 2^{2^n}$$

$$2^n \geq c \cdot 2^n \cdot 2^n$$

$$1 \geq c \cdot 2^n$$

$$\begin{matrix} f(n) = n \\ g(n) = 2^n \end{matrix}$$

Complexity

Classes

$$2^n = O(n^n)$$

$$2^n \leq c \cdot n^n$$

$$f(n) = O(g(n))$$
$$f(n) \leq c \cdot g(n)$$

$$2^n = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdots - n \text{ times}$$

$$n^n = n \cdot n \cdot n \cdot n \cdots n - n \text{ times}$$

$$n=3$$

4, 5, 6

2
2
2

(3) ~~2^n~~

$2^n = O(n^{\sqrt{n}})$
 ~~$2^n \leq C \cdot n^{\sqrt{n}}$~~ ~~False~~

$f(n) \leq C \cdot g(n)$

$\Rightarrow f(n) = O(\underline{g(n)})$

2^n | $n^{\sqrt{n}}$

take log

$\log_2 2^n$

$\log_2 n^{\sqrt{n}}$

$n \times 1$

$\sqrt{n} \log_2 n$

~~$\sqrt{n} \cdot \sqrt{n}$~~

~~$\sqrt{n} \cdot \log_2 n$~~

$\sqrt{n}$

$\log_2 n$

$n^{1/2}$		$\log n$
$\log n^{1/2}$		$\log \log n$
$\frac{1}{2} \log n$		$\log \log n$

$$\log n > \log \log n$$

$$2^n > \frac{n^{\sqrt{n}}}{n}$$

$$2^n > n^{\log n}$$

④

$$2^n = O(n^{\log n})$$

$$2^n \leq c \cdot n^{\log n}$$

False

$$2^n \geq c \cdot n^{\log n}$$

2^n		$n^{\log n}$
$\log 2^n$		$\log n^{\log n}$
$n \log 2$		$\log n \times \log n$

$$\begin{array}{l|l}
 n & (\log n)^2 \\
 \log n & \log(\log n)^2 \\
 \log n & 2 \log \log n
 \end{array}$$

$$\log(n) \log \log(n)$$

Imp points:-

$$\begin{array}{l}
 \rightarrow 1. n! = O(n^n) \\
 \rightarrow 2. \log(n!) = \Theta(n \log n)
 \end{array}$$

remember

$$\begin{array}{l}
 \rightarrow 0 \\
 \rightarrow \infty
 \end{array}$$

after log
22 at 1.

$$\begin{aligned}
 n! &= n(n-1)(n-2)(n-3) \dots \\
 n^n &= n \times n \times n \dots n
 \end{aligned}$$

$$n! \leq n^n$$

2.

$$\log(n!) = \theta(n \log n)$$

$$n! \Downarrow O(\cancel{n^n})$$

take

$$\log n! = O(\log n^n)$$

$$\log n! = O(\underline{n \log n})$$

⑥ ~~if~~ $f(n) = O(f(n/2)) \Rightarrow \underline{\underline{\text{false}}}$

①

$$f(n) = n^2 + 2$$

$$f(n/2) = \left(\frac{n}{2}\right)^2 + 2$$

$$= \frac{n^2}{4} + 2$$

$$\frac{n^2}{2} + 2 \leq c \left(\frac{n^2}{4} + 2 \right)$$

②

$$f(n) = 2^n$$

$$f(n/2) = 2^{n/2}$$

$$f(n) = O(f(n/2))$$

$$2^n \leq c \cdot 2^{n/2}$$

complexity error

$$\cancel{2^{n/2}} \cdot 2^{n/2} \leq c \cdot \cancel{2^{n/2}} \quad \text{X}$$

$$\underline{2^{n/2}} \leq c \cdot \underline{1}$$

⑦

$$f(n) = O\left(\frac{f(n)}{2}\right) \quad \text{True}$$

$$\frac{f(n) - 2^n}{2} = \frac{2^n}{2}$$

$$2^n \leq c \cdot \frac{2^n}{2}$$

complexity $\subset$ truth

